

✓ Inspection of The Results

After all the experiments were ran, it is time to inspect and analyze the outcome. There will be the following parts to that:

- 1. General inspection of constraint properties, excluding coverage and regularity. The subjects of the inspection will be the distributions of the constraints by 1) the type of linguistic information they employ; 2) their target item.
- 2. General inspection of the coverage and regularity measures. At this step, the obtained values will be inspected, which includes the analysis of their distribution and correlation.
- 3. Inspection of item against type regularity values. This inspection gives an idea about the presence of negative outliers within covered items of the constraints.
- 4. Inspection of the extent to which the claims about the constraints made in the literature are supported by the data. That includes manual comparison of the quantifiers specified in the constraint definitions ("regularly", "mostly" etc.) and the obtained item regularity values. We concentrate on the item variant since we want a more balanced estimation not affected by productive outliers.
- 5. Inspection of the extent to which our suggestions to the definitions of not fully supported constraints are supported by the data.

✓ 1. General Inspection of Constraint Properties

Our goal for this step of inspection is to retrieve the following:

- 1. The distribution of the constraints by the type of linguistic information they employ.
- 2. The distribution of the constraints by their target items.

Both properties are retrievable directly from the [master constraints file](#) so we load and access it here.

✓ 1.1. Distribution of Constraints by Linguistic Information Type

Beside the linguistic info type, we will also distinguish between the application type (P2I/L2P).

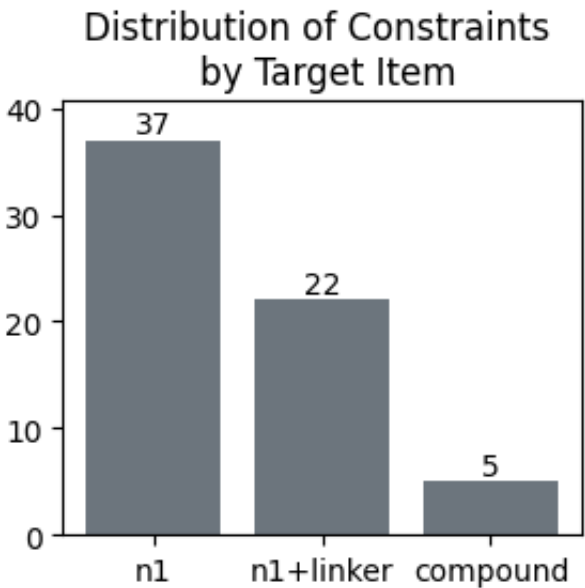
Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.		
	p2l	l2p
derivational	9	2
derivational+morphological	1	1
derivational+morphological+phonological	1	0
derivational+semantic	2	0
lexical	1	4
lexical+morphological	0	1
lexical+morphological+phonological+semantic	0	1
lexical+semantic	1	0
morphological	6	6
morphological+phonological	2	1
morphological+phonological+semantic	3	1
morphological+semantic	8	1
phonological	5	4
semantic	2	0
# P2L and L2P constraints		
(41, 22)		

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

```
morphological    33
derivational     16
phonological     18
semantic         19
lexical          8
dtype: int64
```

1.2. Distribution of Constraint by Target Item

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.



Additionally, we want to know which number of possible items each type has.

Beginnen Sie mit dem Programmieren oder generieren Sie Code mit KI.

```
n1                3613
n1+linker         4868
n2                7210
compound         282767
dtype: int64
```

2. General Inspection of The Coverage and Regularity Measures

Here, we load the general applicability statistics table and retrieve three things:

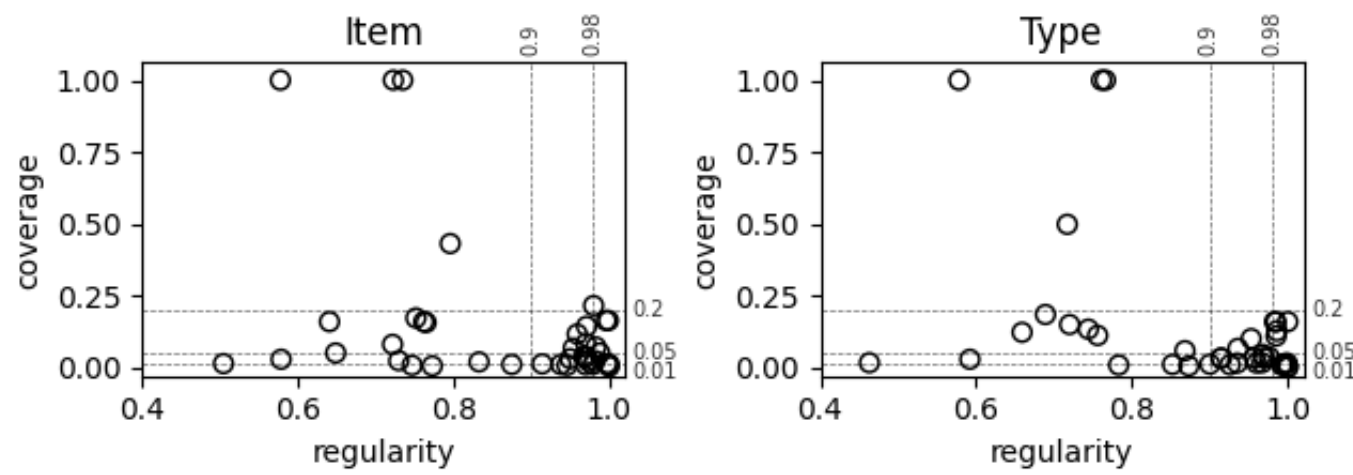
- 1. Scatter plots of coverage against regularity in two variants each (item, type).
 - A shared plot.
 - P2L vs L2P (in this iteration, effectively n1 vs n1+linker).
- 2. Central tendency measures of the distributions (shared, P2L, L2P).
- 3. Qualitative analysis of the values clusters.

2.1. Draw Scatter Plot

Now draw the plots. We start with the shared plot.

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

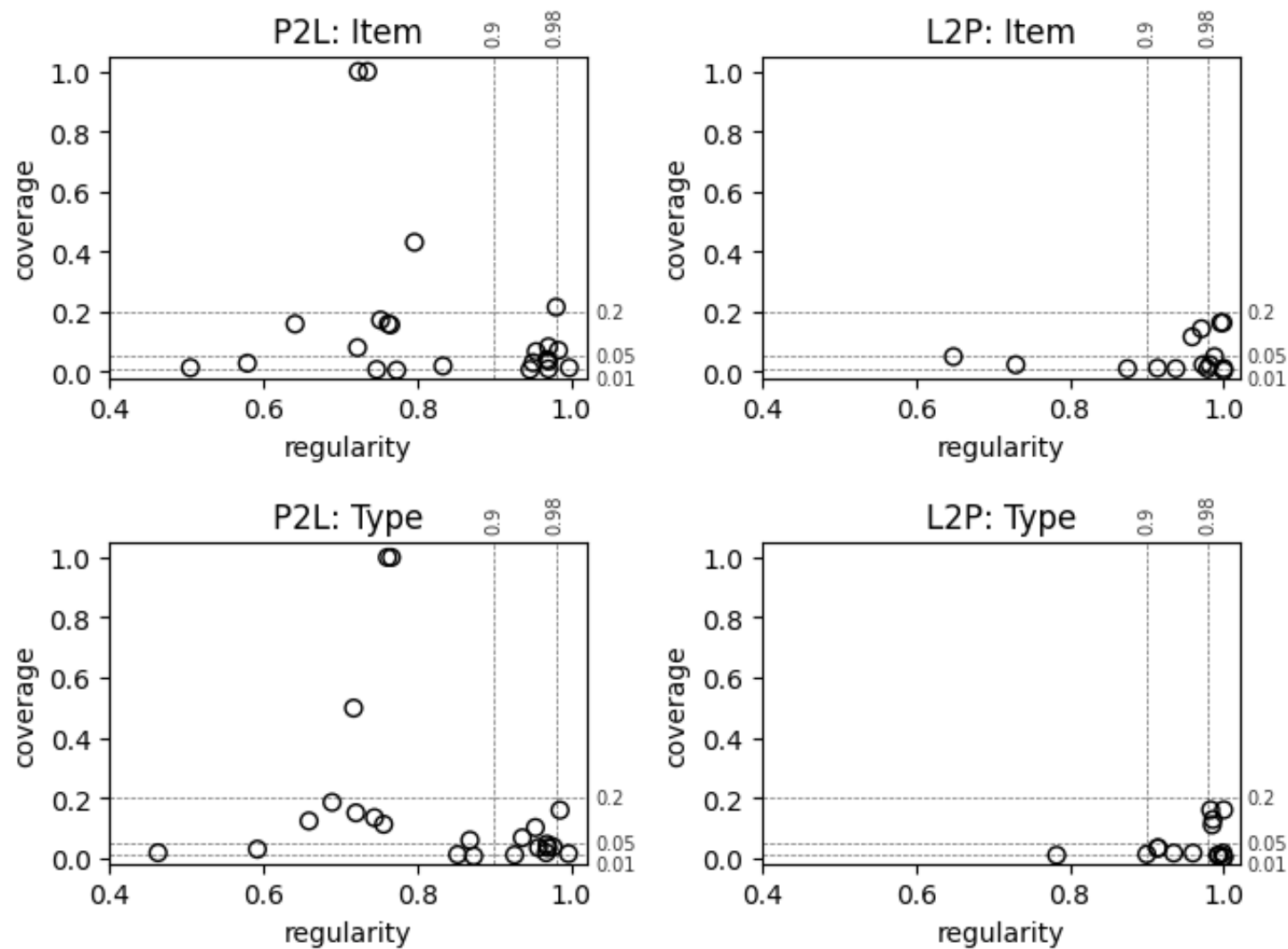
Coverage and Regularity of German Nominal Compounding Constraints



Proceed to the P2L vs L2P plot.

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

Coverage and Regularity of German Nominal Compounding Constraints: P2L vs. L2P



2.2. Central Tendency Measures

We want to obtain min, max, mean, median, and sd of the values in three variants: shared, P2L, L2P.

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

	shared: cvlg_item	shared: cvlg_type	shared: reg_item	shared: reg_type	p21: cvlg_item	p21: cvlg_type	p21: reg_item	p21: reg_type	l2p: cvlg_item	l2p: cvlg_type
min	0.003	0.000	0.505	0.463	0.005	0.005	0.505	0.463	0.003	0.000
max	1.000	1.000	1.000	1.000	1.000	1.000	0.997	0.996	0.163	0.159
mean	0.144	0.142	0.860	0.872	0.165	0.166	0.827	0.830	0.054	0.047
median	0.041	0.033	0.946	0.935	0.067	0.058	0.796	0.868	0.023	0.015
sd	0.264	0.266	0.144	0.141	0.281	0.283	0.144	0.147	0.060	0.059

2.3. Qualitative Analysis of Value Clusters

Now, we check the numbers of different groups of constraints by their coverage and regularity. The groups are chosen by a visual inspection of the plot; the choice aims at a more illustrative representation of the results.

- Many of the coverage values are concentrated in the bottom of the plots so we explore values under 0.2, 0.05, and 0.015.
- There is a slight shift of regularity towards 1.0 so we group values over 0.9 and 0.98.

Many of the coverage values are concentrated in the bottom of the plots.

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

	shared	p21	l2p
cvlg_over_0.2	4	3	0
cvlg_under_0.2	34	19	15
cvlg_under_0.05	19	10	9
cvlg_under_0.01	3	2	1
reg_over_0.9	20	9	11
reg_over_0.98	6	1	5

4. Inspection of The Extent to Which the Claims About the Constraints Made in The Literature Are Supported by The Data

The following steps are to be done for final inspection:

1. Manually inspect the support level of the claims in the empirical data. For each (analyzed) constraint, record the level of support. Since the definitions are mostly vague, base the decision on commonsense judgments. For the support level, we concentrate on the type variant of the measures since we focus on the linguistic side of the question (not real usage). There are 4 levels of support:
 - Supported. Full or very close correspondence between the quantifier and the empirical data. Examples: "regular" vs 0.99, "mostly" vs 0.75.
 - Partially supported. The gap is not critical yet noticeable. Examples: "regular" vs 0.93, "tends" vs 0.6, "sometimes" vs 0.9.
 - Not supported. The discrepancy between the quantifier and the empirical data is too significant to ignore. Examples: "regular" vs 0.85, "tends" vs 0.4.
 - NA: no data available (unreliable or semantic/lexical constraints).

Note: this steps is being conducted in the file with constraints definitions, see `constraints/constraints.yaml`. While it is being now made at the time of this inspection, we cannot reflect this processes here. After this step is done, we pull the results from this file and further inspect them in detail, specifically:

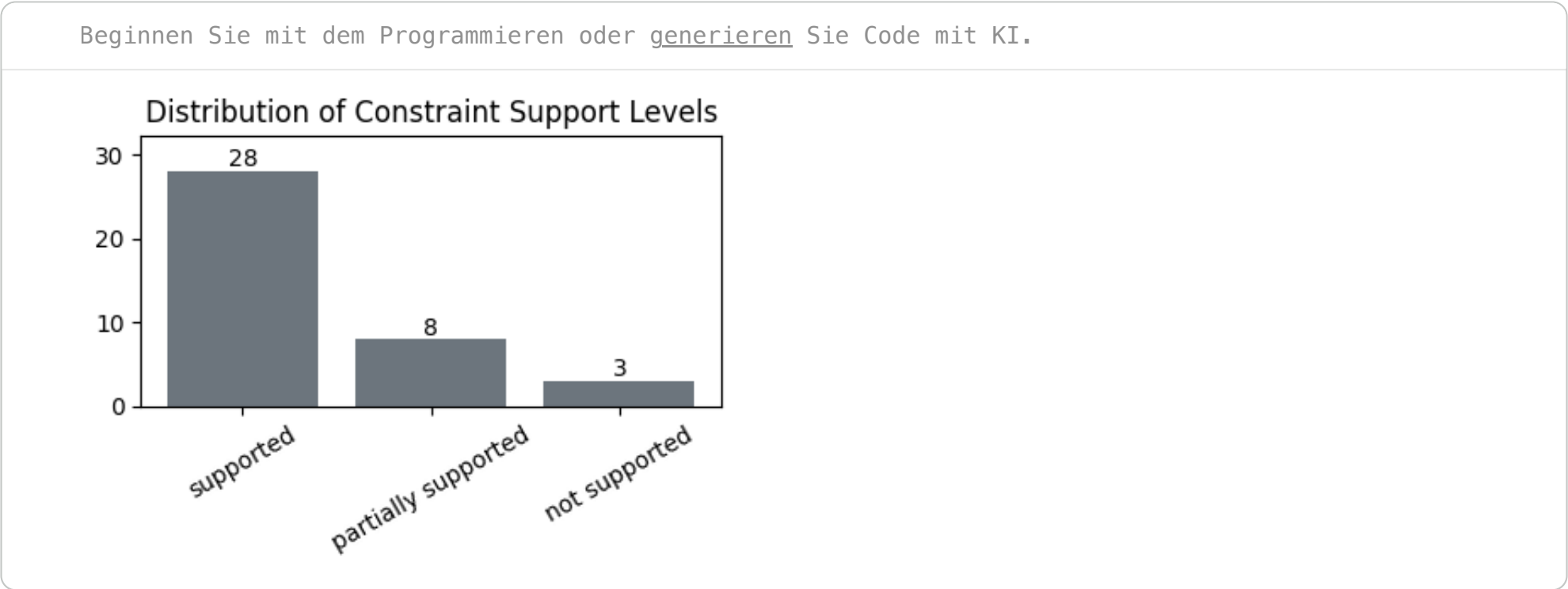
2. Inspect the distribution of the support levels.
3. Manually inspect the constraints that are only partially supported by the data or not supported at all.

4.1. Inspection of The Support Level

Done separately in `constraints/constraints.yaml`.

4.2. Inspection of Support Level Distribution

Now, we read the constraint definition file for inspection.



4.3. Qualitative Inspection of Partially / Not Supported Constraints

Partially supported constraints.

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

```
[ 'p2l:decl_cl:pl:#s-0',  
  'p2l:phon_fin:vow$-0',  
  'p2l:phon_fin:F_stem_#t$-s',  
  'l2p:n',  
  'l2p:en|par',  
  'l2p:e|#stressed_syl',  
  'l2p:er|#stressed_syl',  
  'l2p:es|mono_syl']
```

a few highly productive N1s: 'Karneval', 'Fasching', 'Interim', 'Harem'

p2l:decl_cl:pl:#s-0_item n_bw_item		
comp_gecodb		
karneval_+s_land	karneval	194
karneval_+s_künstler	karneval	194
karneval_+s_luft	karneval	194
fasching_+s_brauchtum	fasching	129
fasching_+s_rede	fasching	129
fasching_+s_auftritt	fasching	129
interim_+s_situation	interim	64
interim_+s_wohnung	interim	64
interim_+s_kapitän	interim	64
harem_+s_hose	harem	11
harem_+s_wächter	harem	11
harem_+s_verschwörung	harem	11
deck_+s_mann	deck	7
deck_+s_fläche	deck	7
deck_+s_balken	deck	7
etat_+s_jahr	etat	1

mostly loanwords ending in a long stressed [e] ('ee') or [i] ('ie'):
'Kaffee', 'Fee', 'Melodie', 'Chemie'
a few examples with 'Schuh' and 'Zeg'

p21:phon_fin:vow\$-0_item n_bw_item		
comp_gecodb		
idee_+n_bringer	idee	118
idee_+n_zentrum	idee	118
idee_+n_fetzen	idee	118
fee_+n_geschichte	fee	39
fee_+n_kreis	fee	39
fee_+n_hügel	fee	39
kalorie_+n_angabe	kalorie	36
kalorie_+n_nachschub	kalorie	36
kalorie_+n_versorgung	kalorie	36
melodie_+n_strom	melodie	18
melodie_+n_auswahl	melodie	18
melodie_+n_reichtum	melodie	18
zeh_+en_schuh	zeh	13
zeh_+en_nagel	zeh	13
zeh_+en_entzündung	zeh	13
galerie_+n_quartier	galerie	9
galerie_+n_dichte	galerie	9
galerie_+n_verband	galerie	9
kategorie_+n_sieger	kategorie	5
kategorie_+n_lehre	kategorie	5
kategorie_+n_bildung	kategorie	5
moschee_+n_streit	moschee	3
moschee_+n_gemeinde	moschee	3
moschee_+n_verband	moschee	3
schuh_+e_kauf	schuh	3
schuh_+e_binde	schuh	3
schuh_+e_schnur	schuh	3
etat_+s_jahr	etat	1
kuh_+=e_dorf	kuh	1
energie_+n_gesetz	energie	1

many highly productive N1 such as 'Umwelt', 'Heimat', 'Antwort'

p2l:phon_fin:F_stem_#t\$-s_item n_bw_item		
comp_gecodb		
jugend_behörde	jugend	583
jugend_gefieder	jugend	583
jugend_rente	jugend	583
umwelt_medizin	umwelt	514
umwelt_komponente	umwelt	514
...	...	...
armut_bekämpfung	armut	2
armut_zeugnis	armut	2
herkunft_staat	herkunft	2
herkunft_land	herkunft	2
andacht_stätte	andacht	1

113 rows x 2 columns

almost all word ending in [ɐ] ('er') such as 'Oper', 'Nummer'

12p:n_item n_bw_item		
comp_gecodb		
oper_+n_gast	oper_+n_	262
oper_+n_unternehmen	oper_+n_	262
oper_+n_tournee	oper_+n_	262
schwester_+n_kleid	schwester_+n_	50
schwester_+n_kampf	schwester_+n_	50
schwester_+n_personal	schwester_+n_	50
nummer_+n_verzeichnis	nummer_+n_	35
nummer_+n_bereich	nummer_+n_	35
nummer_+n_anzeige	nummer_+n_	35
jungfer_+n_bastei	jungfer_+n_	32
jungfer_+n_brunnen	jungfer_+n_	32
jungfer_+n_weiher	jungfer_+n_	32
auster_+n_essen	auster_+n_	26
auster_+n_perle	auster_+n_	26
auster_+n_wasser	auster_+n_	26
wimper_+n_kranz	wimper_+n_	16
wimper_+n_spirale	wimper_+n_	16
wimper_+n_schlag	wimper_+n_	16
zeder_+n_saal	zeder_+n_	12
zeder_+n_baum	zeder_+n_	12
zeder_+n_stamm	zeder_+n_	12
maser_+n_infektion	maser_+n_	10
maser_+n_welle	maser_+n_	10
maser_+n_impfung	maser_+n_	10
kaper_+n_beere	kaper_+n_	6
kaper_+n_ersatz	kaper_+n_	6

kaper_+n_strauch	kaper_+n_	6
feder_+n_hersteller	feder_+n_	5
feder_+n_kleid	feder_+n_	5
feder_+n_werk	feder_+n_	5
elster_+n_weg	elster_+n_	3
elster_+n_nest	elster_+n_	3
elster_+n_paar	elster_+n_	3
ader_+n_verkalkung	ader_+n_	3
ader_+n_paar	ader_+n_	3
ader_+n_geflecht	ader_+n_	3
mittel_+n_kirche	mittel_+n_	2
mittel_+n_dorf	mittel_+n_	2
wetter_+n_straße	wetter_+n_	1
nachbar_+n_land	nachbar_+n_	1
ammer_+n_dorf	ammer_+n_	1
semmel_+n_knödel	semmel_+n_	1
zwiebel_+n_schneide	zwiebel_+n_	1
mauer_+n_straße	mauer_+n_	1
waffel_+n_backe	waffel_+n_	1
windel_+n_wechsel	windel_+n_	1
vokabel_+n_pauke	vokabel_+n_	1
bürger_+n_meister	bürger_+n_	1

a number of unrelated N1

12p:en par_item n_bw_item		
comp_gecodb		
stern_+en_taler	stern_+en_	197
stern_+en_regen	stern_+en_	197
stern_+en_muster	stern_+en_	197
instrument_+en_tausch	instrument_+en_	97
instrument_+en_zubehör	instrument_+en_	97
instrument_+en_begleitung	instrument_+en_	97
storch_+en_treffen	storch_+en_	90
storch_+en_kenner	storch_+en_	90
storch_+en_villa	storch_+en_	90
hahn_+en_bach	hahn_+en_	35
hahn_+en_busch	hahn_+en_	35
hahn_+en_königin	hahn_+en_	35
länge_+n_wachstum	länge_+n_	22
länge_+n_messung	länge_+n_	22
länge_+n_zuwachs	länge_+n_	22
dorn_+en_hecke	dorn_+en_	19
dorn_+en_wald	dorn_+en_	19
dorn_+en_ranke	dorn_+en_	19
isotop_+en_trennung	isotop_+en_	12

isotop_+en_signatur	isotop_+en_	12
isotop_+en_muster	isotop_+en_	12
schelm_+en_markt	schelm_+en_	9
schelm_+en_graben	schelm_+en_	9
schelm_+en_kopf	schelm_+en_	9
kaper_+n_ersatz	kaper_+n_	6
kaper_+n_apfel	kaper_+n_	6
kaper_+n_strauch	kaper_+n_	6
star_+en_abwehr	star_+en_	5
star_+en_nest	star_+en_	5
star_+en_kasten	star_+en_	5
glaube_+n_können	glaube_+n_	3
glaube_+n_bekenntnis	glaube_+n_	3
glaube_+n_gemeinschaft	glaube_+n_	3
hitze_+n_hof	hitze_+n_	2
hitze_+n_dorf	hitze_+n_	2
mittel_+n_dorf	mittel_+n_	2
mittel_+n_kirche	mittel_+n_	2
bürger_+n_meister	bürger_+n_	1
treue_+n_feld	treue_+n_	1
hirse_+n_mühle	hirse_+n_	1
derivat_+en_handel	derivat_+en_	1
ruhe_+n_stand	ruhe_+n_	1
wetter_+n_straße	wetter_+n_	1
reue_+n_tal	reue_+n_	1

# prefixed derivatives from 'Halt'		
12p:e #stressed_syl_item n_bw_item		
comp_gecodb		
anhalt_+e_kelle	anhalt_+e_	15
anhalt_+e_stelle	anhalt_+e_	15
anhalt_+e_gebot	anhalt_+e_	15
inhalt_+eproduktion	inhalt_+e_	3
inhalt_+eangebot	inhalt_+e_	3
inhalt_+engeschäft	inhalt_+e_	3
vorjahr_+esieger	vorjahr_+e_	1

'Mitglied'

12p:er #stressed_syl_item n_bw_item		
comp_gecodb		
mitglied_+=er_betreuung	mitglied_+=er_	163
mitglied_+=er_konto	mitglied_+=er_	163
mitglied_+=er_partei	mitglied_+=er_	163
altertum_+=er_sammlung	altertum_+=er_	3
altertum_+=er_behörde	altertum_+=er_	3
altertum_+=er_verwaltung	altertum_+=er_	3
vorbild_+=er_sammlung	vorbild_+=er_	1

prefixed derivatives from a few covered N1 + 'Gesetz'

12p:es mono_syl_item n_bw_item		
comp_gecodb		
gesetz_+es_regelung	gesetz_+es_	179
gesetz_+es_verstoß	gesetz_+es_	179
gesetz_+es_kenntnis	gesetz_+es_	179
vorjahr_+es_kurs	vorjahr_+es_	108
vorjahr_+es_majestät	vorjahr_+es_	108
vorjahr_+es_rang	vorjahr_+es_	108
vortag_+es_gewinn	vortag_+es_	11
vortag_+es_sieger	vortag_+es_	11
vortag_+es_geschäft	vortag_+es_	11
bestand_+es_aufnahme	bestand_+es_	2
bestand_+es_führung	bestand_+es_	2
gesang_+es_kunst	gesang_+es_	1

Not supported constraints.

Beginnen Sie mit dem Programmieren oder [generieren](#) Sie Code mit KI.

```
['p2l:drv:sfx:e_pl_sfx-0',
 'p2l:drv:sfx:deverb_#en$-s',
 'p2l:phon_fin:schwa$-en|f-en']
```

a number of N1 ending in -at: 'Internat', 'Magistrat' etc.

p2l:drv:sfx:e_pl_sfx-0_item n_bw_item		
comp_gecodb		
internat_+s_leitung	internat	52
internat_+s_konzept	internat	52
internat_+s_jahr	internat	52
dekanat_+s_verband	dekanat	39
dekanat_+s_gemeinde	dekanat	39
dekanat_+s_leitung	dekanat	39
magistrat_+s_vertreter	magistrat	31
magistrat_+s_mitglied	magistrat	31
magistrat_+s_verfassung	magistrat	31
plagiat_+s_opfer	plagiat	21
plagiat_+s_prüfer	plagiat	21
plagiat_+s_klage	plagiat	21
derivat_+e_börse	derivat	18
derivat_+e_anleger	derivat	18
derivat_+e_strategie	derivat	18
rektorat_+s_übergabe	rektorat	16
rektorat_+s_kollegium	rektorat	16
rektorat_+s_kirche	rektorat	16
patronat_+s_recht	patronat	11
patronat_+s_vertrag	patronat	11
patronat_+s_mittel	patronat	11
quadrat_+e_stadt	quadrat	3
quadrat_+e_meter	quadrat	3
quadrat_+e_städter	quadrat	3
implantat_+e_hersteller	implantat	1
personal_+s_chef	personal	1

* Around a third: zero-linker variants
of doubtful cases N1-(s)-N2 whose frequency
is considerably lower than that of the corresponding
s-linker variants: 'Lebenstil' (vs 'Lebensstil')
* A few zero-linker variants with a higher frequency than
the s-linker variant: 'Lebengeschäft' (vs 'Lebensgeschäft')
* Around half: compounds that don't have s-linker variants:
'Bebengebiet', 'Bratensoße', 'Hustentropfen'

p2l:drv:sfx:deverb_#en\$-s_item n_bw_item		
comp_gecodb		
essen_auswahl	essen	42
essen_entzug	essen	42
essen_plan	essen	42
leben_lauf	leben	26
leben_sparte	leben	26
leben_dorf	leben	26
braten_ansatz	braten	26
braten_rezept	braten	26

braten_test	braten	20
braten_brot	braten	26
husten_bonbon	husten	18
husten_anfall	husten	18
husten_krampf	husten	18
beben_gebiet	beben	13
beben_gefahr	beben	13
beben_schaden	beben	13
wissen_vermittlung	wissen	12
wissen_durst	wissen	12
wissen_bach	wissen	12
vorhaben_plan	vorhaben	10
vorhaben_genehmigung	vorhaben	10
vorhaben_katalog	vorhaben	10
unternehmen_scheck	unternehmen	7
unternehmen_gruppe	unternehmen	7
unternehmen_bestuerung	unternehmen	7
einkommen_situation	einkommen	6
einkommen_bestuerung	einkommen	6
gewissen_grund	gewissen	6
gewissen_frage	gewissen	6
gewissen_konflikt	gewissen	6
schrecken_szenarium	schrecken	6
schrecken_stein	schrecken	6
schrecken_sturm	schrecken	6
einkommen_steigerung	einkommen	6
leiden_zeit	leiden	5
leiden_frost	leiden	5
vertrauen_verhaeltnis	vertrauen	5
vertrauen_verlust	vertrauen	5
vertrauen_beweis	vertrauen	5
leiden_schaf	leiden	5
vermoegen_wert	vermoegen	4
vermoegen_scheck	vermoegen	4
vermoegen_schaden	vermoegen	4

a great number of deadjectival and deverbal N1:
'Reise', 'Probe', 'Wärme'; also unrelated 'Gebäude'

p2l:phon_fin:schwa\$-en f-en_item n_bw_item		
comp_gecodb		
reise_kasse	reise	510
reise_zurückhaltung	reise	510
reise_tochter	reise	510
liebe_+s_sucher	liebe	434
liebe_+s_blume	liebe	434
...	...	...
rute_möller	rute	1
rinde_rohr	rinde	1
länge_feld	länge	1
pfeife_raucher	pfeife	1
printe_mann	printe	1

736 rows x 2 columns

5. Inspection of The Extent to Which Our Constraint Correction Suggestions Are Supported by The Data

stats of the corrected constraints

n_appl_items corrected_quantifier resulting_cvg_item resulting_cvg_type resultin				
constr_id				
p2l:drv:sfx:e_pl_sfx-0	3614	regularly	0.005	0.004
p2l:drv:sfx:deverb_#en\$-s	3614	mostly	0.005	0.009
p2l:phon_fin:vow\$-0	3614	regularly	0.026	0.024
l2p:n	4870	all	0.116	0.109
l2p:el#stressed_syl	4870	all	0.023	0.008
l2p:erl#stressed_syl	4870	all	0.010	0.015
...	...	..	...	...

if all constraints are now supported
all(sl == "supported" for sl in sls_after_corr)

True